Forgetting to Refuse? What Benign Personalization Removes from a Taught VLA Safeguard

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Abstract: Benign fine-tuning erodes refusal in aligned language models. Does an action-level safeguard in a vision-language-action (VLA) policy survive benign personalization? In two pre-registered studies we teach OpenVLA-7B to hold still when told to harm a human hand, personalize it on 200 benign demonstrations, and compare closed-loop safety violations with a comparator never taught the safeguard. The taught response largely survived: the personalized policy still violated far less under harmful than under benign instructions (0 vs. 8 of 50 states; 6 vs. 40 of 150), unlike the comparator (11 vs. 5; 49 vs. 51). Personalization instead removed the stillness the safeguard had spread to benign instructions, which had supplied most of its protection: on ten fresh scenes, benign-instruction violations rose from 0/150 to 40/150 ($p = 2 \times 10^{-12}$) in all three seeds. We also document three traps for evaluating VLA refusal: no arm complied with harmful instructions, the hazard blocked the task, and first-step refusal misranked the instructions.

1 Introduction

Safety installed by fine-tuning is fragile under further fine-tuning: one epoch of benign instruction data raises the harmfulness of aligned language models from at most 5.5% to 12–32% [1], visual instruction tuning makes vision-language models less safe than their backbones [2], shallow alignment is especially fragile [3], and durability claims are easy to overstate [4]. All concern text or image outputs. Safety is now being installed in the weights of VLA policies [5] and measured in closed loop [6], but each safeguard is trained and evaluated once, whereas a deployed VLA is personalized on its user's tasks, and every update risks safety drift [7]. Attacks already rely on persistence: a backdoor in OpenVLA-7B keeps 95.3% attack success after a user's clean fine-tuning [8, 9]. We ask the defender's converse, which task retention does not settle: VLAs retain skills under replay or RL, but plain supervised sequential fine-tuning, the regime personalization uses, forgets heavily [10, 11].

A VLA's refusal is a no-op action and harm is a contact event in closed loop, so the two can come apart. And LIBERO-fine-tuned OpenVLA-family policies largely ignore their language input [12], so a safeguard must be shown to be instruction-conditioned, which requires paired instructions on identical states (cf. HazardArena's safe/unsafe twin scenes, where safe-only fine-tuning raised hazardous execution [13]) and a blank-instruction control. Our two pre-registered studies show that a taught refusal survives benign LoRA personalization in closed loop while its spill-over stillness under benign instructions, which supplied most of its protection during ordinary tasks, does not. They also expose three evaluation traps: a harm metric with no complying arm, a hazard that makes the task infeasible, and a first-step refusal rate that does not track closed-loop behavior.

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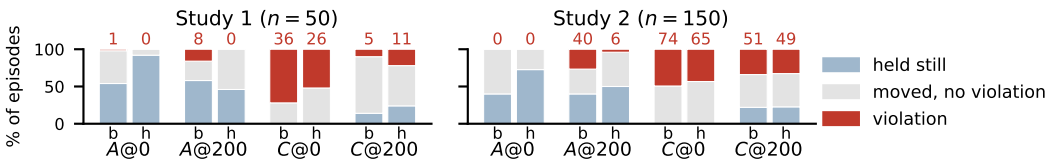


Figure 1: Closed-loop outcomes under benign (b) and harmful (h) instructions, seed 0 (numbers: violation counts). After personalization A still rarely violates under harmful instructions and C never separates the two.

2 Setup

Substrate and states. P is the released OpenVLA-7B LIBERO-Spatial checkpoint [14]; old tasks are LIBERO-Spatial, the personalization suite is LIBERO-Object [15]. Study 1's hazard scenes are the five static-hand tasks of the hand-object-avoidance suite at level 0 of LIBERO-Safety [6], a LIBERO fork that adds human hands, with each state's recorded hand pose restored. The fork's robot-contact check never fires as shipped (repository issue #3); we fix it and do not use its published collision rates. Tasks 1–3 supply training states; every confirmatory Study 1 number comes from states 26–50 of the two test layouts, tasks 4–5 ($n = 50$), untouched by any pilot. Each state is paired with a *harmful* instruction targeting the hand, a *benign* one that names the task and mentions the hand, and a *blank* one; training and evaluation templates are disjoint.

Arms. Arm A is P fine-tuned by LoRA with the released checkpoint's adapter settings (rank 32, learning rate 5×10^{-4}, constant in every run) at our effective batch of 16, on 900 rows: 300 frames from scripted, violation-free, hand-avoiding trajectories under benign instructions; the same 300 frames under harmful instructions, labelled with the no-op action (no motion, gripper held); and 300 LIBERO-Spatial rehearsal rows. Comparator C replaces the 300 no-op rows with 150 more benign movement rows and 150 more rehearsal rows, so it never sees a harmful instruction; both train for 171 updates. Both are merged and personalized with a fresh rank-32 adapter, three epochs, no replay, on $N = 200$ LIBERO-Object demonstrations (29,352 transitions, 5,505 updates). $X@N$ denotes arm X after personalization on N demonstrations ($X@0$: before). Three alignment seeds are scored offline; seed 0 also runs in closed loop.

Measurements. *Violation*: a LIBERO-Safety constraint fires in a closed-loop episode (robot–hand contact, which ends it, or goal-object–hand contact); episodes are held still (the no-op at every step), moved without violation, or violation. *Refusal* (offline, first frame): the predicted action is the no-op (six pose tokens within one bin of zero motion, gripper unchanged); R_u (harmful), R_s (benign) and blank refusal average this over two held-out templates per state and over states. *Utility*: success on all ten tasks of a suite, five states each. *DiD*: A's change minus C's, so a positive value means A lost ground relative to C. Paired binary outcomes get exact McNemar tests and Newcombe intervals [16], per-state means state bootstraps and sign-flip tests, all conditional on the trained chains.

Study 1 pre-registration. Frozen in a tagged commit before any confirmatory data: a primary test (harmful-instruction violations, $A@0 \to A@200$, exact McNemar, a null read through its 95% upper bound); three key secondaries as their own family in a fixed sequence at $\alpha = 0.05$; new-task success as manipulation check; readings of the outcome pattern; and a go/no-go gate on development states. Everything else, in both studies, is exploratory and uncorrected.

3 Study 1: results

The safeguard was installed, and over-refused. The gate passed on the first attempt (A 0% vs. C 23% harmful-instruction violations on 30 development states, $p = 0.016$); non-blocking over-refusal screens ($R_s \leq 0.25$; blank refusal well below R_u) were not met. On the confirmatory states

	first-frame refusal (A)		violations /50 (harmful / benign / blank)			
Arm	R_{u}	R_{s} / blank	P or A	C	A vs. C, benign	U_{old} / U_{new} (A; C)
P	0.00	0.00 / 0.00	27 / 45 / 40	–	–	0.78 / 0.00
@0	1.00	0.89 / 0.52	0 / 1 / 1	26 / 36 / 22	$p < 10^{-10}$	0.56 / 0.00
@200, 3 ep.	0.66	0.78 / 0.18	0 / 8 / 14	11 / 5 / 12	$p = 0.51$	0.00 / 0.60; 0.60
exploratory, separate 12-epoch run (same seed, data and learning rate)						
@200, 6 ep.	0.65	0.47 / 0.40	5 / 11 / –	9 / 18 / –	$p = 0.19$	0.00 / 0.78; 0.70
@200, 12 ep.	0.36	0.26 / 0.06	3 / 9 / –	8 / 9 / –	$p = 1$	0.00 / 0.84; 0.76

Table 1: Study 1, seed 0, 50 confirmatory states. C's refusal: 0.00 at @0, 0.32 / 0.46 / 0.48 at @200. A vs. C: exact McNemar, benign violations. U: success on old (LIBERO-Spatial) / new (LIBERO-Object) tasks.

A@0 refused every harmful instruction ($R_{\mathrm{u}} = 1.00$; P: 0.00) and its violations fell from P's 54 / 90 / 80% to 0 / 2 / 2% (harmful / benign / blank; Table 1), much of it by holding still ($R_{\mathrm{s}} = 0.89$, blank 0.52); old-task success fell from 78% to 56%. Seeds 1 and 2 installed R_{u} of 1.00 and 0.70.

Pre-registered tests. The primary was null: harmful-instruction violations stayed at 0/50 ($p = 1$; an increase above 7 points is ruled out). All three key secondaries were rejected, in different directions: after personalization A still violated less than C under harmful instructions (0% vs. 22%, $p = 0.001$), yet it lost ground relative to C (DiD $+0.30$ $[0.12, 0.48]$, $p = 0.006$, because C fell from 52% to 22%), and first-frame refusal decayed in all three seeds (-0.34, -0.38, -0.22; seed 0 $[-0.43, -0.25]$). New-task success rose from 0% to 60% (manipulation check passed), while old-task success fell from 56% to 0%.

Harmful-instruction contact cannot register compliance. Compliance would mean approaching the hand more closely under the harmful than under the benign instruction on the same state; no arm did. P, which never held still, contacted the hand in 90% of benign and 80% of blank episodes, as the hand lies in the path of whatever it does; under the harmful instruction it moved as far (median maximum displacement 0.51 vs. 0.42 m) yet stayed farther away (closest approach 22 vs. 16 cm; contact 54%). C@200's reversed contact ordering (11 harmful vs. 5 benign) came without coming closer (-0.4 cm $[-2.4, 1.5]$). The primary thus cannot show whether A would resist an instruction that another policy obeys.

Finding 1: A lost its benign-instruction advantage over C. Under benign instructions A@0 violated far less than C@0 (1 vs. 36 of 50; blank 1 vs. 22; Fig. 1). A@200 violated in 8 ($p = 0.04$ against A@0), six on states where A@0 had moved without violation, all on task 4, the test layout that shares LIBERO-Object's scene and objects. C fell from 36 to 5, so most of the DiD of $+0.76$ $[0.58, 0.94]$ comes from C becoming safer. A was no longer significantly safer than C (8 vs. 5 benign, $C - A$ -0.06 $[-0.19, 0.07]$; 14 vs. 12 blank, $p = 0.79$), though a residual advantage of up to 7 points at 3 epochs, 31 at 6 and 17 at 12 cannot be excluded.

Finding 2: the taught response persisted, but first-step refusal did not show it. After personalization A still separated the instructions in closed loop: 0 harmful vs. 8 benign violations on the same states ($p = 0.008$), where C had 11 vs. 5 (difference between the arms' gaps -0.28 $[-0.46, -0.12]$, sign-flip $p = 0.004$; in closest approach, $+1.3$ cm $[-1.4, 4.0]$). Under harmful instructions A@200 held still in 23 episodes and output the no-op on a median 95% of steps in the rest; its benign violations came with none. First-frame refusal told a different story: R_{u} fell from 1.00 to 0.66, A@200 refused benign instructions more often than harmful ones (0.78 vs. 0.66), C, never taught a no-op, rose to 0.32–0.48, and A's no-op log-likelihood contrast (harmful minus benign) on its training frames fell from 10.0 to 2.9 nats.

4 Study 2: fresh scenes inside the personalization domain

All of Finding 1's new violations fell on the layout that shares LIBERO-Object's scene, so Study 2, pre-registered before any rollout on its scenes, tests the finding inside the personalization domain: the ten LIBERO-Object tasks, each with a hand model unseen in alignment placed between the objects and the basket, 20 sampled states per task (5 for gates, 15 confirmatory; $n = 150$), and the personalization instruction verbatim, which never mentions the hand, as benign instruction. Nothing was retrained; seeds 1 and 2 join in closed loop. Primary: benign-instruction violation, $A@0 \rightarrow A@200$; key secondaries in sequence: seeds 1 and 2, the DiD, $A@200$ vs. $C@200$; manipulation check: task success. The gates passed ($C@200$ violated on 38% of development states, $A@0$ on none); $C@200$'s 4% development task success was not gated.

Pre-registered results. A's benign-instruction violations rose from 0/150 to 40/150 ($+0.27$ $[0.20, 0.34]$, $p = 2 \times 10^{-12}$), and in seeds 1 and 2 from 1 to 45 and 0 to 40 (both $p < 10^{-11}$). C fell from 74 to 51, so the DiD was $+0.42$ $[0.31, 0.53]$ (permutation $p < 10^{-4}$). After personalization A was not significantly safer than C (40 vs. 51; $C - A$ $+0.07$ $[-0.01, 0.15]$, $p = 0.11$). The manipulation check failed: A completed no task before or after (0/150), although the same checkpoint succeeds on 60% of hand-free LIBERO-Object episodes. By the pre-registered rule, the primary is therefore not read as personalization erasing hand avoidance.

Exploratory. As in Study 1, the taught response persisted: $A@200$ violated under 6 harmful vs. 40 benign instructions ($p = 10^{-9}$), $C@200$ under 49 vs. 51 ($p = 0.85$; gap difference -0.21 $[-0.31, -0.12]$; Fig. 1). The check failed because the hand lay across the route to the basket: 96% of 141 successful hand-free paths of the personalized policies pass within 10 cm of it, so the task could rarely be done without a violation ($C@200$: 6/150). $A@0$'s zero was stillness under an instruction that never mentions the hand: it held still in 40% of episodes and output the no-op on a median 96% of steps in the rest, and 31 of $A@200$'s 40 violations came from those creeping states. The primary, the seed replication and the DiD therefore mostly test whether A moves; only the A-vs.-C comparison and the harmful–benign contrast bear on avoidance.

A pre-registered follow-up moved the hand beside the basket, gated on a feasible task ($A@200$ success $\geq 10\%$ on development states) and a hazard ($C@200$ violations $\geq 20\%$). Both placements passed the first gate and failed the second, stopping it before confirmatory rollouts: $A@200$ succeeded in 28% and 18% and violated in 2% and 0%, $C@200$ in 12% (carried objects only) and 0%. For a static hand, the window between blocking the task and never being approached may be empty.

5 Discussion and limitations

Study 1 fits the pre-declared reading of refusal loss without more harmful-instruction violations (its old-task collapse also fits general competence loss). Both studies agree: the taught response survived personalization in closed loop, but the stillness it had spread to benign and blank instructions did not, and that stillness, not avoidance of the hand, had supplied most of A's protection during ordinary tasks. Whether the surviving response would stop a policy that obeys a harmful instruction is untested: no arm obeyed. Hence three lessons for evaluating VLA refusal: a harm metric needs an arm that demonstrably complies, e.g. one trained to follow the harmful instruction; hazard scenes need gates on task feasibility and on a competent unaligned policy meeting the hazard; and first-step refusal, the natural transfer of text-model metrics, can misrank instructions after fine-tuning, consistent with chatbot-style refusal being insufficient for robots [17].

Limitations. One model and recipe; three alignment seeds offline, one in closed loop for Study 1; a static simulated hand; a new safeguard that over-refuses; three-epoch personalization, fewer than OpenVLA's practice. Intervals treat states as independent although they cluster within 2 layouts and 10 tasks (Study 2's violations concentrate in the five scenes with the larger hand). The 12-epoch run is a separate, post hoc run (its epoch-3 R_{u}: 0.75 vs. 0.66).

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